

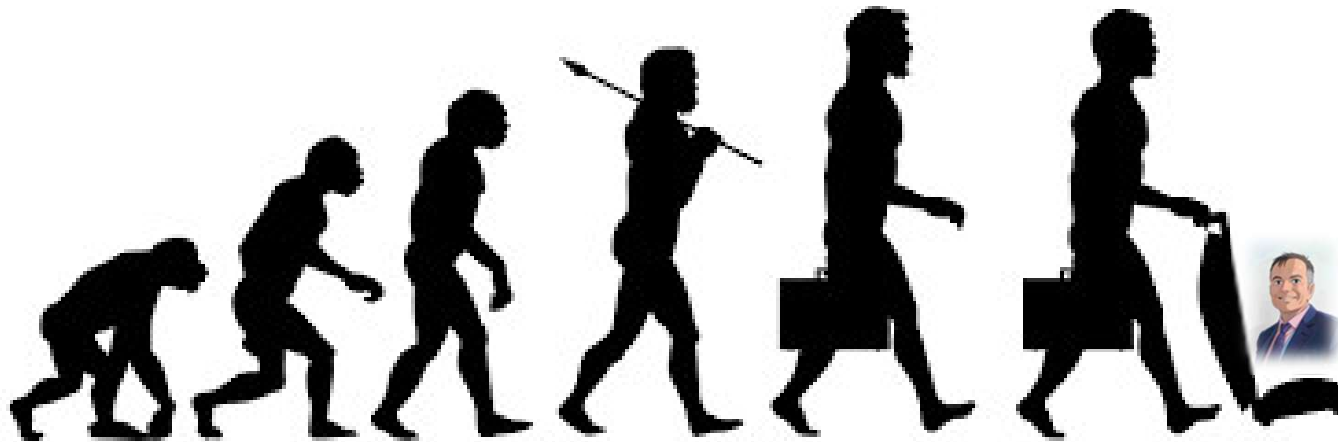


Data Management e Data Strategy



12 November 2025

Who I am



The evolution from Data User to Data Evangelist



Mario Vellella is a member of the DAMA Italy Scientific Committee and Director Irion's Domain Advisory.

He holds a PhD and has over 25 years of experience in the finance sector, having previously worked as a **researcher, consultant and manager in the area of risk management, as well as a collaborator** with the Department of Economics and Management at Sapienza University of Rome.

Author of national and international scientific publications, Mario has extensive expertise as a Domain Expert in Data Governance and integrated risk management, gained both as a consultant to multinational companies and within the Poste Italiane Group.

In particular, he was Chief Data Officer for BancoPosta, responsible for the Integrated Risk Analysis and Model Validation Office

Agenda

- ❑ *DAMA in a nutshell*
- ❑ *Data Driven Company*
- ❑ *Data Integration as a balance point between resilience, meaning, and democracy*
- ❑ *Data Governance: a new corporate governance process*
- ❑ Q&A





DAMA in a nutshell



DAMA - Data Management Association - About us

DAMA INTERNATIONAL

Data Association Management (DAMA) was founded in the 1980s to address data management issues.

DAMA is vendor-independent, **technology-independent**, and **method-independent**. It is the most important global community for advancing and promoting knowledge and skills in the field of **data management**.

DAMA ITALY

DAMA Italia is the officially affiliated, active, and recognized Italian chapter of the International Data Management Association since November 2018 (<https://dama-italy.org/>).

The DAMA International Timeline



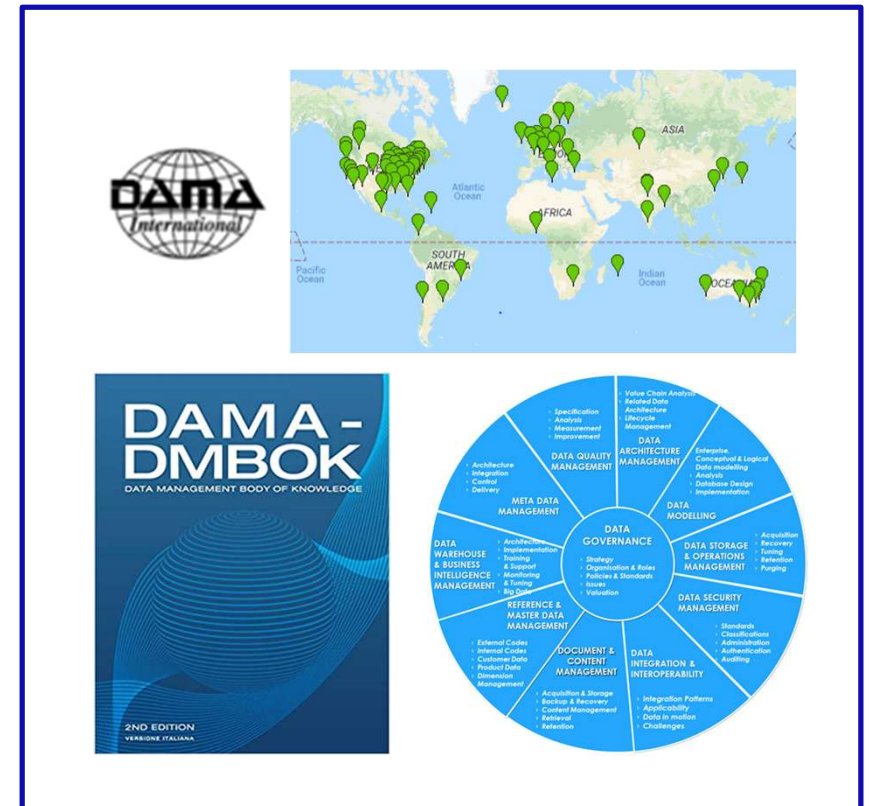
WELCOME TO DAMA-Italy

Il Chapter Italiano dell'Associazione DAMA International



Data Management Association (DAMA) - Assets

- ❑ DAMA brand
- ❑ The Community
- ❑ Data Management Body of Knowledge (DMBoK)
- ❑ Glossary of Terms
- ❑ “The DAMA Wheel”
- ❑ CDMP certification



Data management – what is it?

What is **Data Management (DM)**?

It is a set of **processes, systems, activities, and functions** aimed at the overall **management** of data that are part of the information system, according to their life cycle.

DM is a **cross-cutting approach** that goes beyond the information system (technological sphere) and involves the entire organization at all levels.



DAMA - Data Management Body of Knowledge (DMBoK) - What is it?

The **Data Management Body of Knowledge (DAMA-DMBOK)** is a guide developed by **DAMA International** that defines principles, standards, and best practices for data management and governance. It provides a useful reference **framework and fundamental guidelines** and examples for addressing current challenges in data management. **DMBOK®2** is the second official edition.

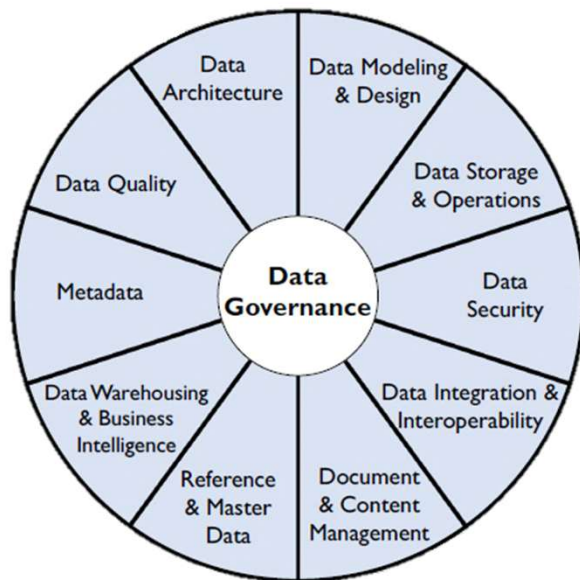
Contents of the DMBoK

- **Data Management standards** and how to apply these principles within organizations, in different functional areas
- a **functional framework** for the implementation of **Data Management**, including widely used practices, methods, and techniques, functions, roles, results, and metrics
- a **common vocabulary** for data management concepts, serving as a basis for best practices for data management professionals

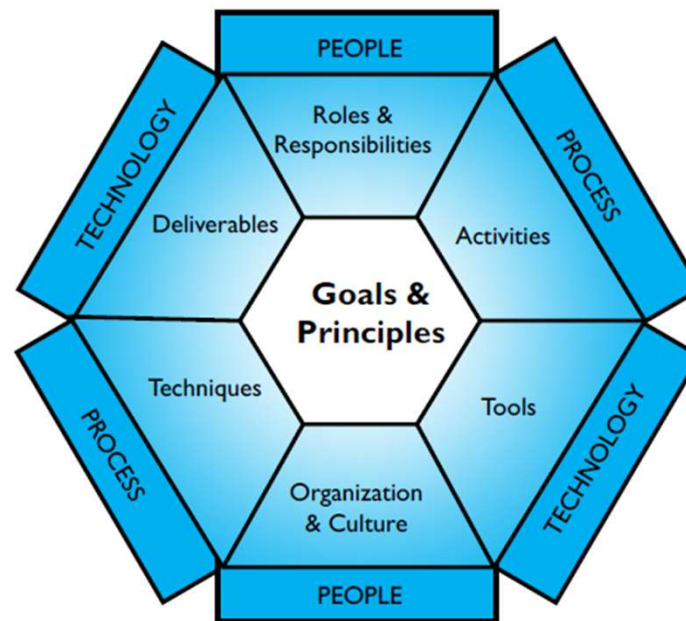
The DAMA Wheel (knowledge area)



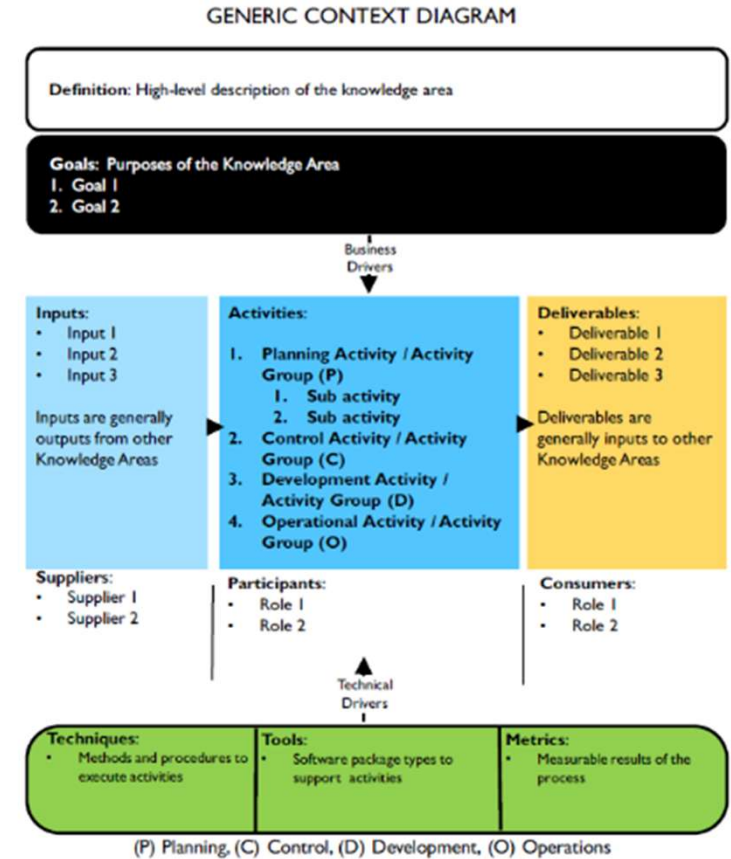
DMBOK – Frameworks



DAMA WHEEL



Environmental Factors hexagon



Knowledge Area Context Diagram (SIPOC)

Certified Data Management Professional (CDMP) – Professional Pathway

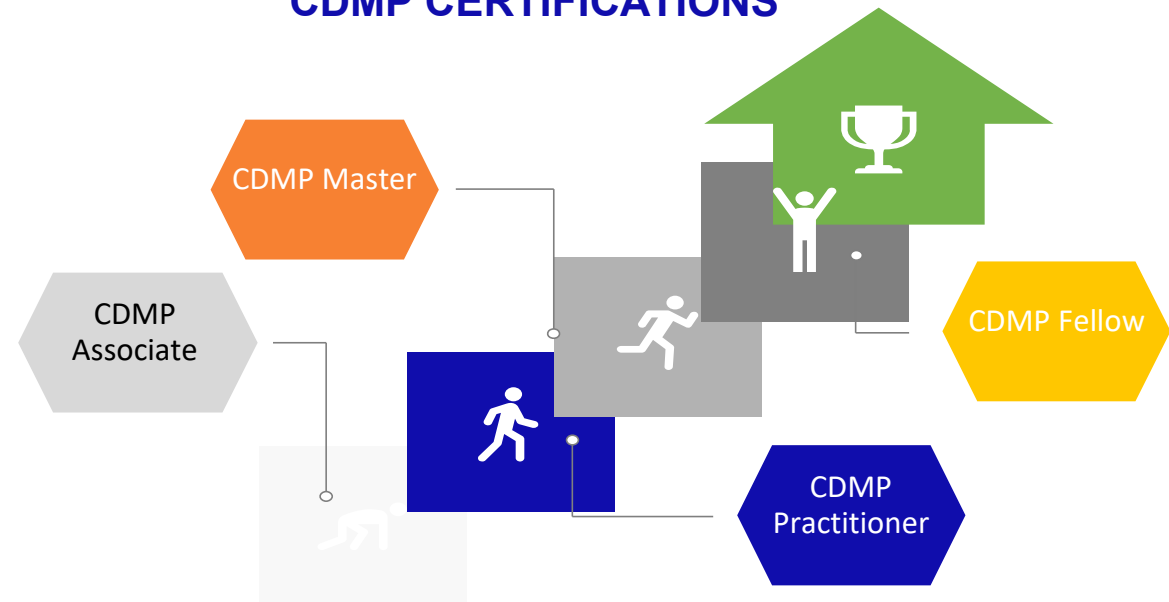
What is **CDMP**?

It is an internationally recognized certification issued by **DAMA International** through the **Data Management Association International**, which certifies professional skills in data management.

CDMP professional path:

- ✓ Training
- ✓ Certification exam
- ✓ Certification levels
- ✓ Continuous professional development

CDMP CERTIFICATIONS



Benefits of CDMP:

- ☐ Global professional recognition
- ☐ Improved career opportunities in data management
- ☐ Greater ability to manage complex data-related projects and processes

- ☐ The certification is valid for three years and must be renewed through the recognition of credits (CDU).
- ☐ The certification is individual (for *professionals*); however, DAMA Italy is currently defining a **certification** method **at the company level**

DAMA Italy: Association membership (professionals)

Individual **Standard Membership:**

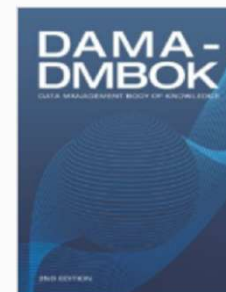
- Access to the reserved area of the **DAMA Italy** website, where you can receive information on the association's activities and events
- Informative and educational materials on Data Management (webinars, abstracts, case studies, etc.)
- Discounts on participation in seminars, events, and conferences organized by the association
- Access to the **DAMA Italy Job Platform** (job board)
- Opportunities to meet with Data and Information Management professionals and speakers
- Access to the Data Management Italia Professional Forum

50 euros/year
(for professionals)



Individual **Plus Membership:**

- All services already included in Individual Standard Membership
- Access to the electronic version of the Data Management Body of Knowledge (DMBoK®2)
- Direct support for the association's activities



95 euros/year
(for professionals)

Memberships are individual and cannot be transferred to third parties. They are valid for one year.

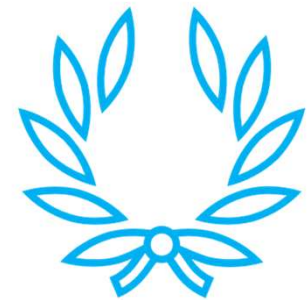
Dama Italy Annual Convention (DIAC)

Annually, DAMA Italy holds its convention (**DAMA Italy Annual Convention – DIAC**) in November, as a crowning achievement of the associative year and a moment of networking and discussion with speakers of great value.



Trip to Turin - Data for Goods

In the context of DIAC, the **DAMA Italy Award** is held, **i.e. the awarding of the Best university theses focused on Data Management.**



To be “DATA DRIVEN”



Why become Data Driven?





The data driven Organization

Challenges

- > Competitors
- > Complexity
- > Speed
- > Compliance
- > Ethics
- > Sustainability
- > ...

Data as an asset

- Volume ■ Ambiguity ■ Security ■ Reliability
- Heterogeneity ■ Value extraction

Data Driven Organisation

Enabling Factors - Modern Data Management

Agile Data Management

- DataOps

Data Culture

- Adaptive Data Governance
- Data Valuation

Active Metadata Management

- augmented data management capabilities
- Augmented Data Quality

Architecture & Design Concepts

- Data Fabric
- Data Mesh ...

Data Driven Behaviour

- ⚙ Managers and 'data literate' professionals manage:
 - ✓ strategies
 - ✓ decisions
 - ✓ Operational Actions
- ⚙ supported by known, governed, guaranteed data
- ⚙ effectively, proactively, punctually and sustainably.

Being Data Driven: informed decisions

What does this mean?

It means having a data-driven approach, to **make** increasingly **informed decisions** based on **objective facts** rather than personal feelings.

Data-driven companies are those that consider data management a **strategic pillar** of their business.

What data is needed?

All **critical data, crucial** to the core business of the company in question.

The **preliminary work** involves:

- identify the important areas of business processes
- find ways to quantify and measure them

How do you become one?

It means transforming data into concrete actions, thanks to:

- **tools** that enable usability and analysis
- clear and comprehensive **processes**
- a widespread **culture**: having data literacy skills to interpret and use data in an informed manner.

Only in this way can critical data be transformed into value and sustainable competitive advantage.

Being Data Driven: A Competitive Advantage - Examples

“An effective data strategy can drive innovation. Ignoring it can cost you your survival.”

Success stories

1. Netflix – Personalization through data

- **Data Strategy:** Collection and analysis of viewing data, user behavior, preferences.
- **Competitive Advantage:** Recommendation algorithms; content production based on tastes.
- **Result:** Increase in subscriptions, reduction in churn, leadership in streaming.

2. Amazon – Supply chain & Customer Experience

- **Data Strategy:** Predictive purchase analysis, logistics optimization, dynamic pricing.
- **Competitive Advantage:** Fast deliveries, seamless customer journey, personalized recommendations.
- **Result:** Continuous growth and eCommerce dominance.

Failure stories

✗ 1. Blockbuster – Ignoring data and changing consumption patterns

- **Lack of Data Strategy:** Failure to analyze signs of change in consumer habits towards digital and on-demand.
- **Missed opportunity:** refusal to buy Netflix, underestimating its data-driven and streaming model
- **Consequence:** an obsolete business model (physical rental), while competitors innovated
- **Result:** Filed for bankruptcy in 2010.

✗ 2. Kodak – Ignoring digital evolution

- **Lack of data strategy:** Failed to leverage market data in the transition to digital.
- **Consequence:** Delay in innovation, underestimation of digital technology.
- **Result:** Failure in 2012, after decades of leadership.

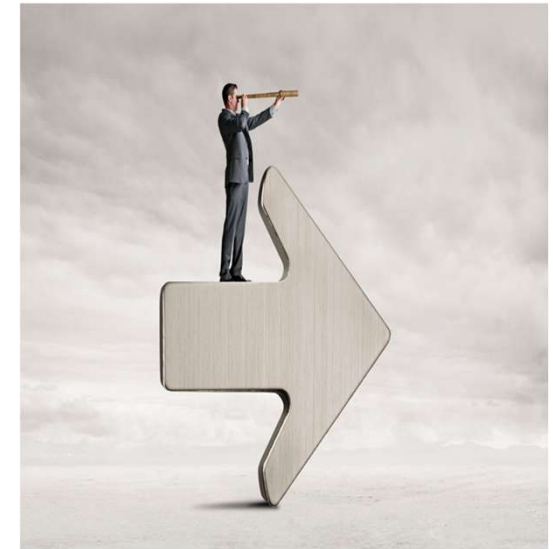
Being data-driven requires a data strategy

What is strategy

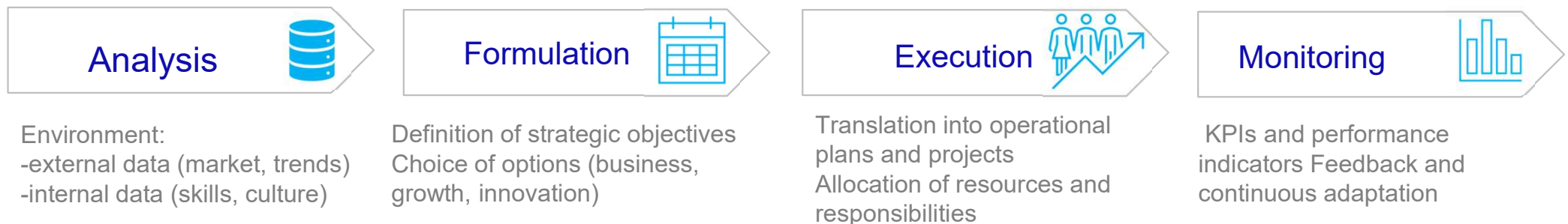
A set of long-term **choices** and **directions** to ensure **competitive advantage** and **value** creation.

Objectives

- Providing an overview of the company
- Direct resources towards key priorities
- Respond to appropriate market challenges and opportunities
- Comply with increasingly complex data regulations



Process implementation



Data Strategy in practice... what is it?



Data Strategy

A data strategy outlines **the principles** that should govern data management within an organisation, translating them into **strategic objectives** that must be pursued through a series of **areas of intervention** in order to achieve and enable one or more business targets.



Strategic Objectives

Trust, Common Semantics, Protection, Interoperability, Value and Quantify, Sharing...

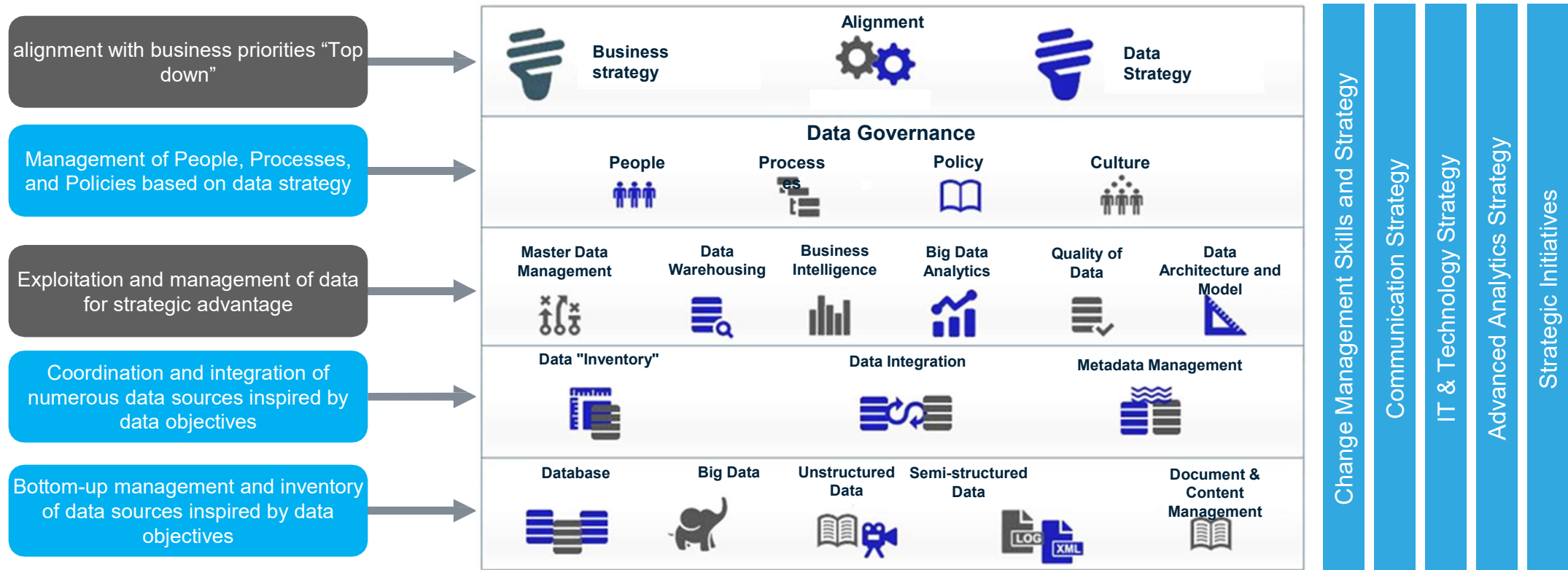


Areas of intervention

Technology, People, Organisation, Regulation.

Implementing a Data Strategy

A successful **data strategy** must be linked to business objectives with organizational and technological solutions



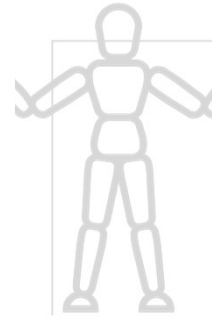
Data Governance e Metadata management”



Agenda

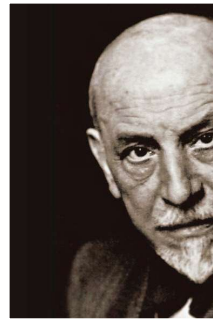


What does it mean to govern data?



Define a DG framework

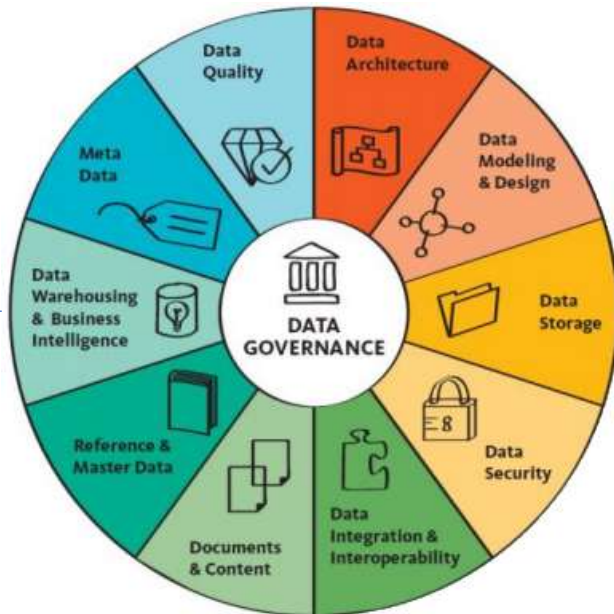
- Organizational model;
- Process model;
- Methodological model;



Governance, metadata management, two disciplines that are closely linked

- Metadata, definitions and basic concepts
- Metadata management

Introduction – What is Governance?



Data Governance is the central element of the DAMA framework.

Its position in the DAMA wheel reflects the enabling role that this management perspective plays in the practical application of the principles and practices outlined in the other areas of knowledge.

Each management perspective focuses on a specific field of action, defining "what" needs to be done,

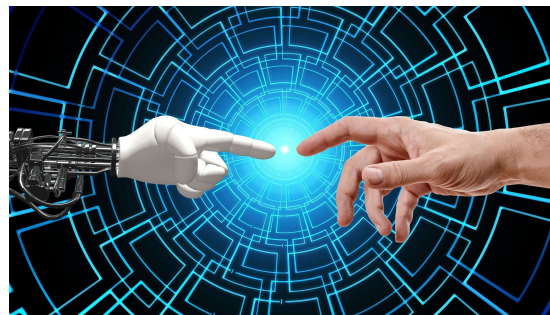
while Data Governance aims to indicate 'how' to do it and with the contribution of 'who'.

"the exercise of authority and control (planning, monitoring, and enforcement) over the management of information assets."

Introduction – What is Governance?



Data governance is both a technical and organizational discipline, and achieving it requires a combination of **data management fundamentals** aligned with **the business organization** and **stakeholder involvement**.



This discipline therefore has the following characteristics:

- It involves multiple skills (**functional, business, organizational, technical, architectural**);
- It is a **guiding** discipline for every business initiative;
- **Communication and leadership skills** are required to implement it.

Why DG - The Data-Driven Company



To manage **people**, a digital company needs an **HR function** and processes that manage recruitment, skills, and retention.



To **manage IT environment**, a digital initiative needs an IT function and processes that ensure technology investments support business aspirations.



To manage its **Processes**, a company needs process management to guide the design and implementation of automated and manual processes.

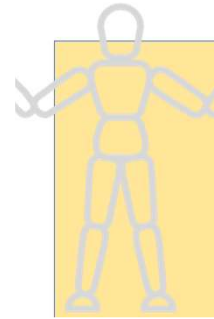


To manage your **data**, you need a data and process management function to **develop and contextualize your data assets in order to support your strategy**.

Agenda

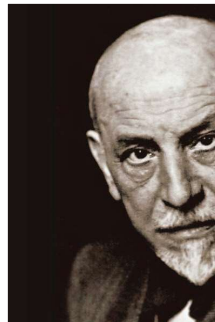


What does it mean to govern data?



Define a DG framework

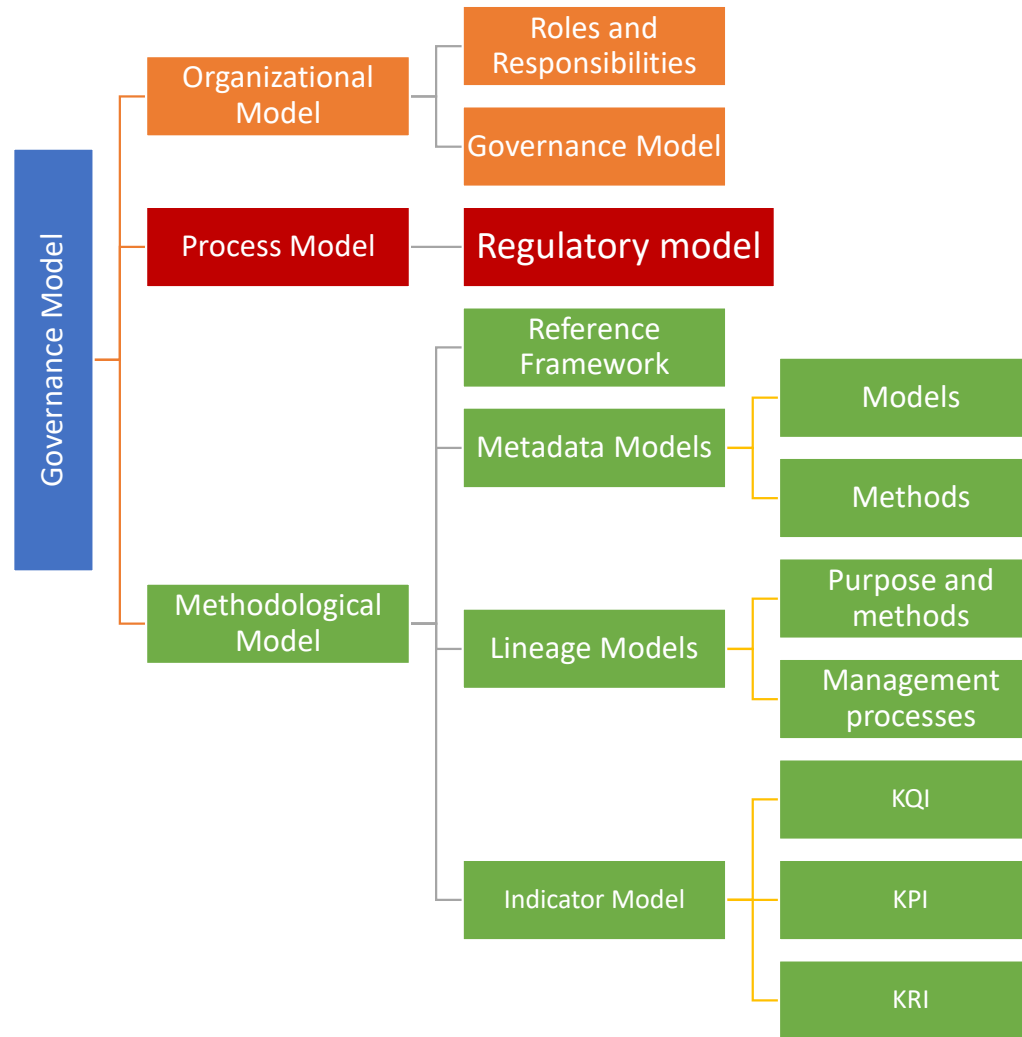
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Defining a Governance Framework



Pills of Organization

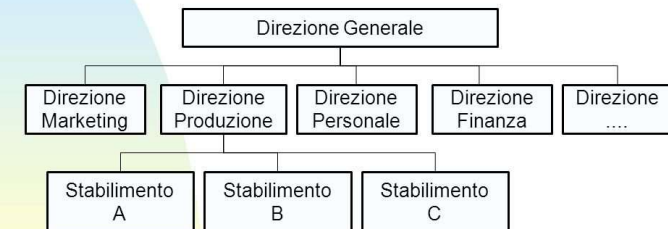
There are several ways to organize a company, here are some examples

Hierarchical model: it is based on the **clear separation between the operational nucleus**, which is the basis of the company, and **the management**. It is the simplest model and is therefore applied to small businesses characterized by a simple workflow and where decisions are made by a few people.



IL MODELLO FUNZIONALE

Functional model: in this case, between the top and the operational department, there is an **intermediate level that deals with coordination functions**, but without real decision-making power. The company is divided into **functional areas according to the activities carried out**. This model is rather rigid and makes it impossible to participate across production processes.



Pills of Organization



Divisional model: it differs from the previous model in the subdivision criteria, **no longer based on functions** but on the basis of variables such as **processes or types of customers**.

Each segment is then **organized internally following the functional model**, simplifying the communication processes between the functions of the same division.

ORGANIGRAMMA DIVISIONALE



Governance models 1/3

Centralized

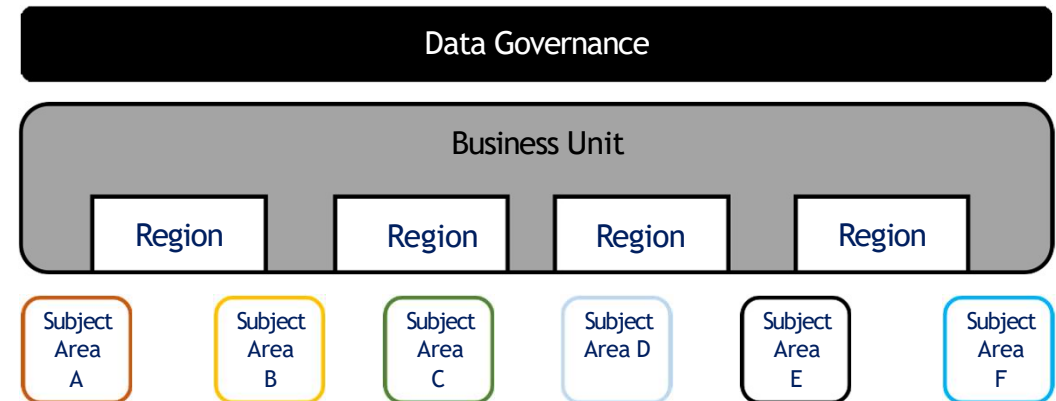
The data governance organization oversees all activities in all subject areas.



- It provides a consistent view of information.
- During the implementation phases, it enables the rapid development of uniform metadata practices.
- In a start up project could be the best configuration

Organizational model

Governance model



- There is a lack of knowledge of individual data domains and their intended uses.
- Highly specialized & technical skills are required.
- Risks being self-referential
- Does not allow for the participation of the actors involved in the process

Governance models 2/3

Decentralized

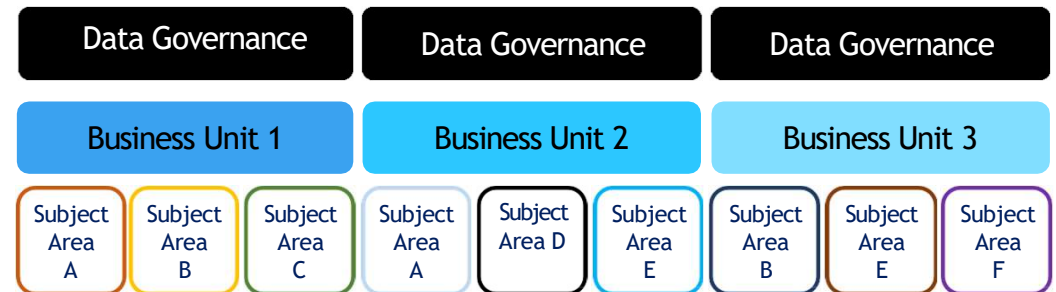
- The same Data Governance operating model is adopted in every business unit.



- Encourages participation and accountability of individual BUs
- Enables the company to be involved in data contextualization activities
- Leverages the vertical knowledge of individual businesses
- Enables the bottom-up model's needs and requirements to be traced



- could create **differences in the approaches actually adopted** in the various BUs. Therefore, overlap in technological choices and inefficiencies.
- May generate **data "silos"** with different rules depending on the BU
- The data culture **could be diversified from BU to BU**, focusing only on BU objectives and not on the overall strategy



Government Models 3/3

Organizational model

Governance model



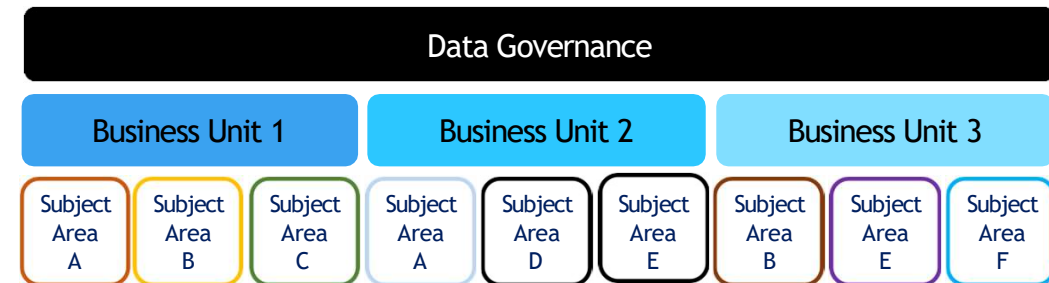
Federated

- A governance function that delegates data governance activities to the business units by issuing consistent methodological guidelines and standards



Global policies ...

(e.g., security, protection, minimum metadata set, ownership attribution rules, interoperability standards, quality criteria)



- In a mature context, it can ensure that the metadata phases are aligned with individual areas of knowledge
- It can generate efficiency, especially in situations where specialists and those with good knowledge of data management have already been identified and are operational.
- It allows for greater independence in choosing the operating model within individual BUs
- Requires a high level of technical & Business expertise
- Risks being self-referential
- Requires a very solid regulatory framework and strict monitoring processes to ensure its implementation

Roles & responsibilities 1/3

Organizational model

Roles and responsibilities



Chief Data Officer (CDO): coordination, support, and guidance point for the entire Group in the process of enhancing the value of its information assets.



Data Owner (DO): Responsible for data within their area of competence. Their objective is to enhance and make these assets available to the organization within the limits established by current regulations.



Enterprise Data Steward and Data Steward The Data Steward is an expert support figure in data management, or a particular asset, who has technical, legal, and ethical expertise in a specific area of competence.



Data User/Consumer (DC): The end user of the data or information resulting from the aggregation/transformation of data.

Roles and responsibilities 2/3

Organizational model

Roles and responsibilities



Data Architect (DA): Responsible for defining a data architecture that facilitates the application of the Data Governance operating model. May assume the role of platform manager if a data platform is adopted.



Data Manager or Technical Owner (DM): Responsible for data from an IT perspective, monitors availability, tracks presence and origin, reliability, and supervises quality together with the Data Owner.



Data Custodian (DC): Responsible for the practical implementation of Data Governance policies. Manages the technology and infrastructure necessary for data security and management, including data protection and storage.



Chief Information Security Officer (CISO): The data security officer plays a crucial role in protecting an organization's digital assets, ensuring that they are managed securely and in compliance with applicable regulations.

Roles and responsibilities 3/3

Organizational Model

Roles and Responsibilities



Governance and Policy Area

Guidelines
Data Strategy

Board

Executive Data Council

Top Management

Coordination and Method Area

Methods, models, and support

CDO/DGO

CDO Dashboard
Quality and KPI

Area of Competence and Responsibility for Data

**Data Owners
and Data Stewards**

Knowledge
Assessment,
monitoring, and
recovery plans

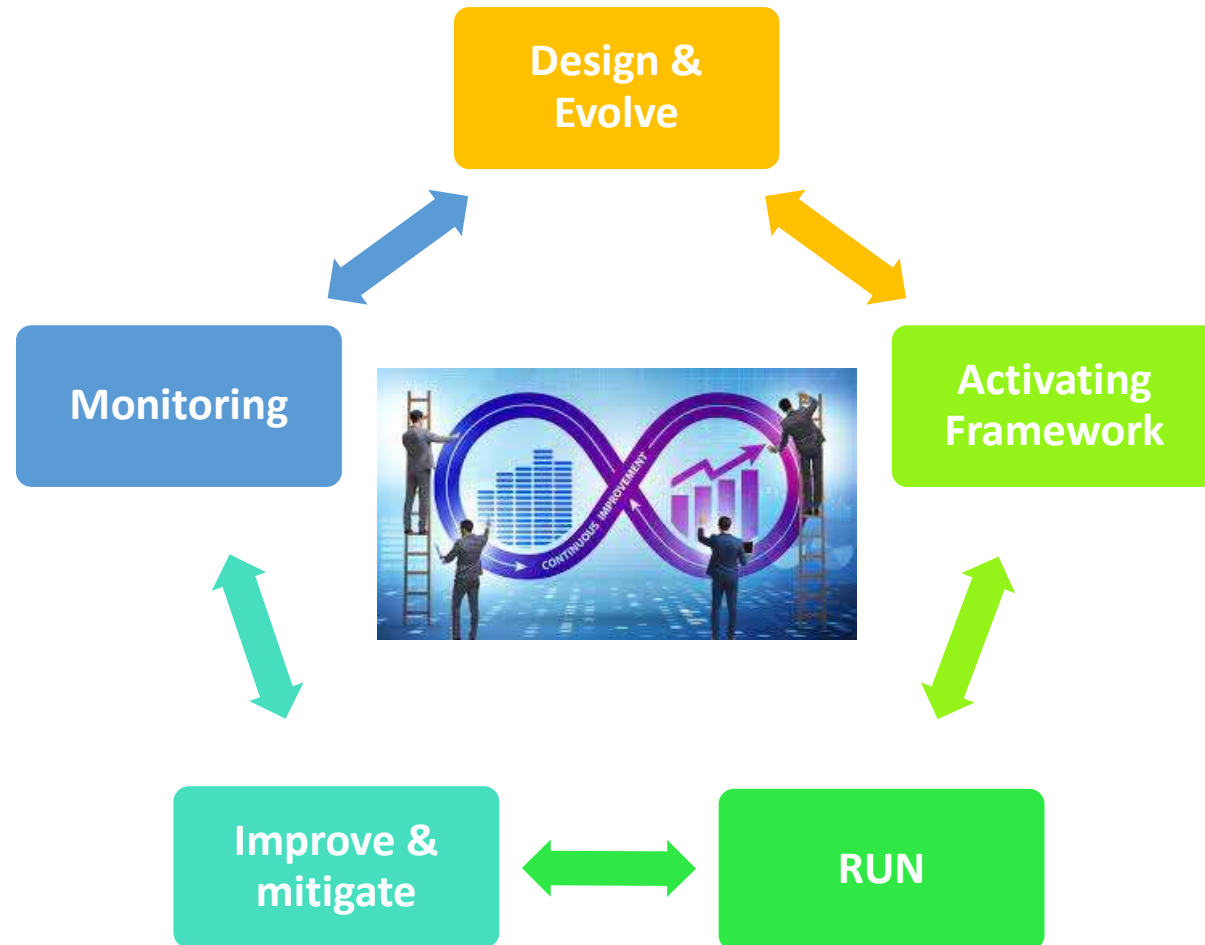
Data Users, Data Consumers

Area of knowledge and Support

**CISO, Data Manager, Data Custodian
Technical Owner, Data Architect**

The process model

Modello dei processi

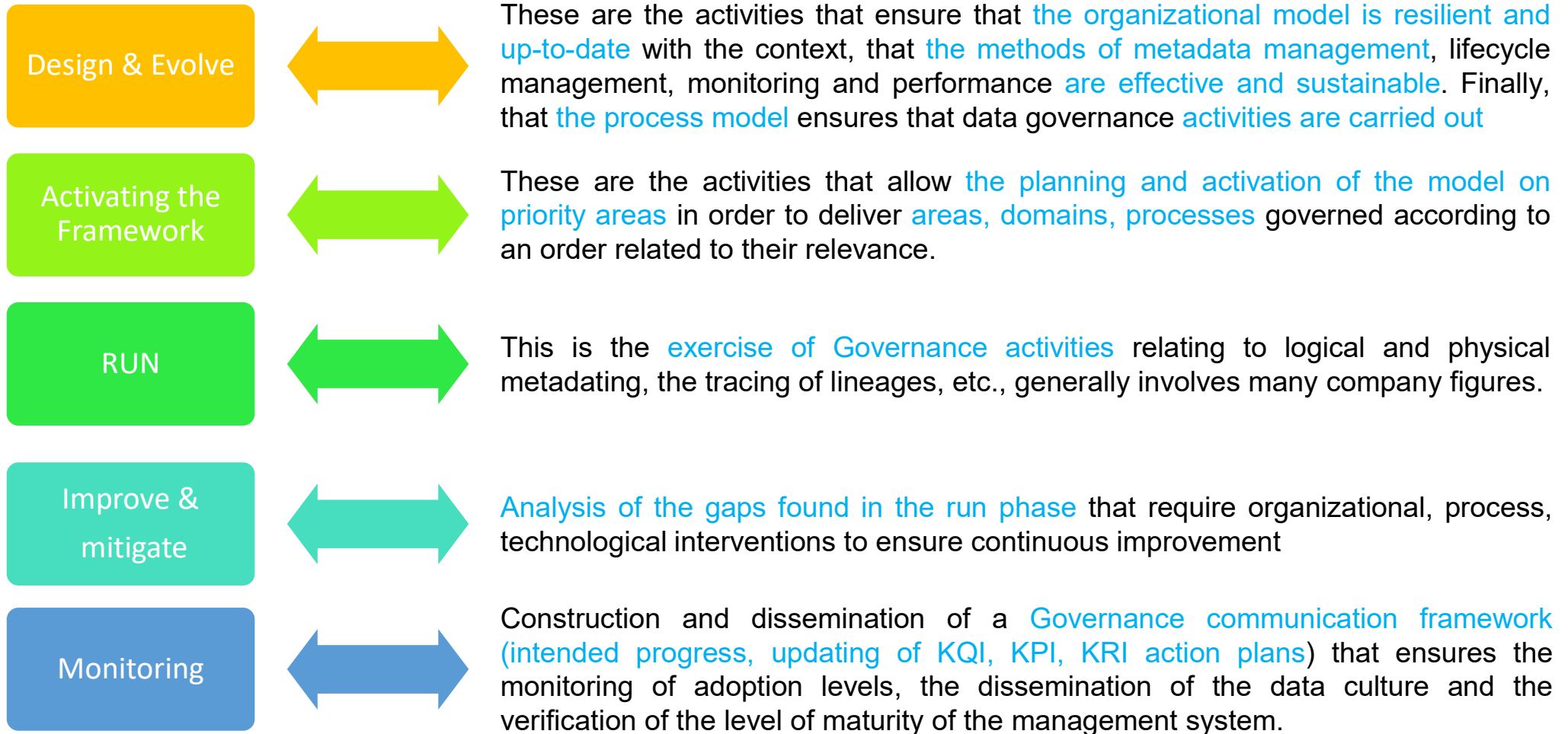


Il modello dei processi

Modello dei processi



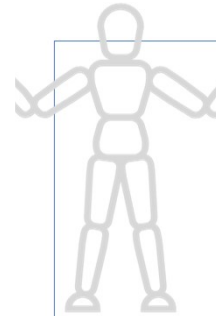
Descrizione



Agenda

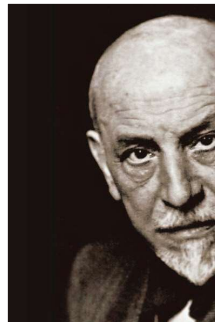


What does it mean to govern data?



Define a DG framework

- Organizational model;
- Process model;
- Methodological model;

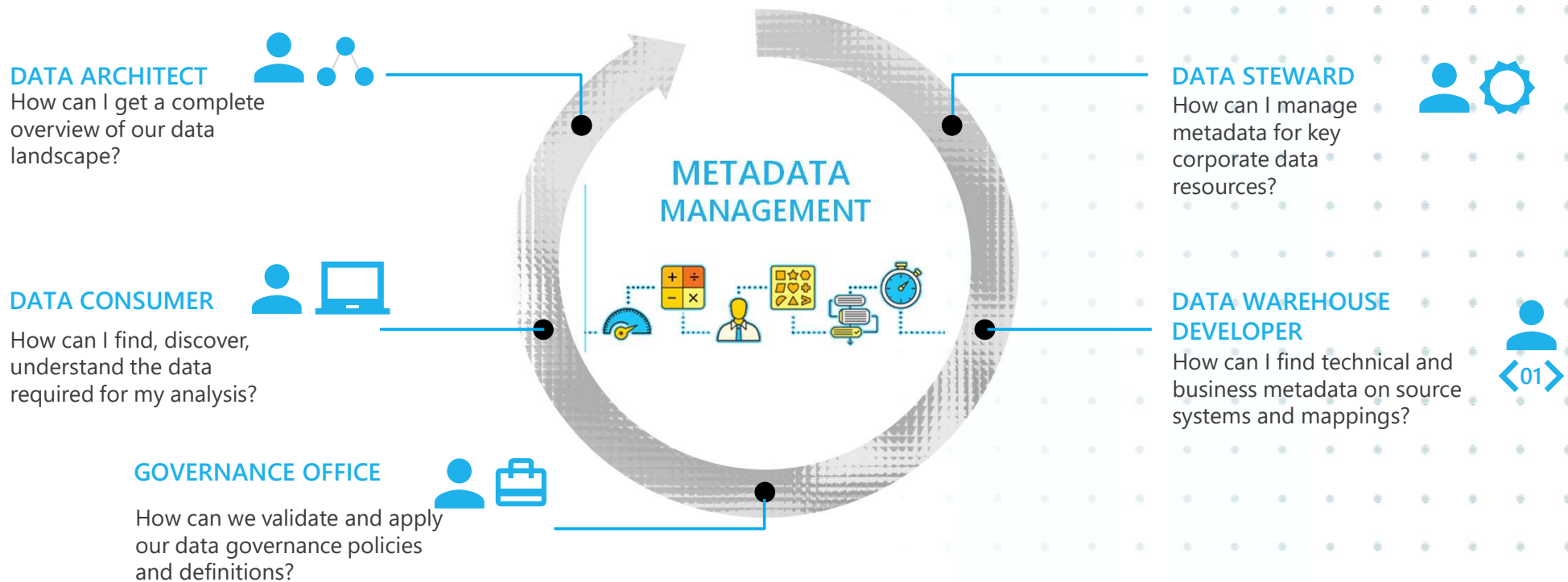


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Metadata are critical for Data Management



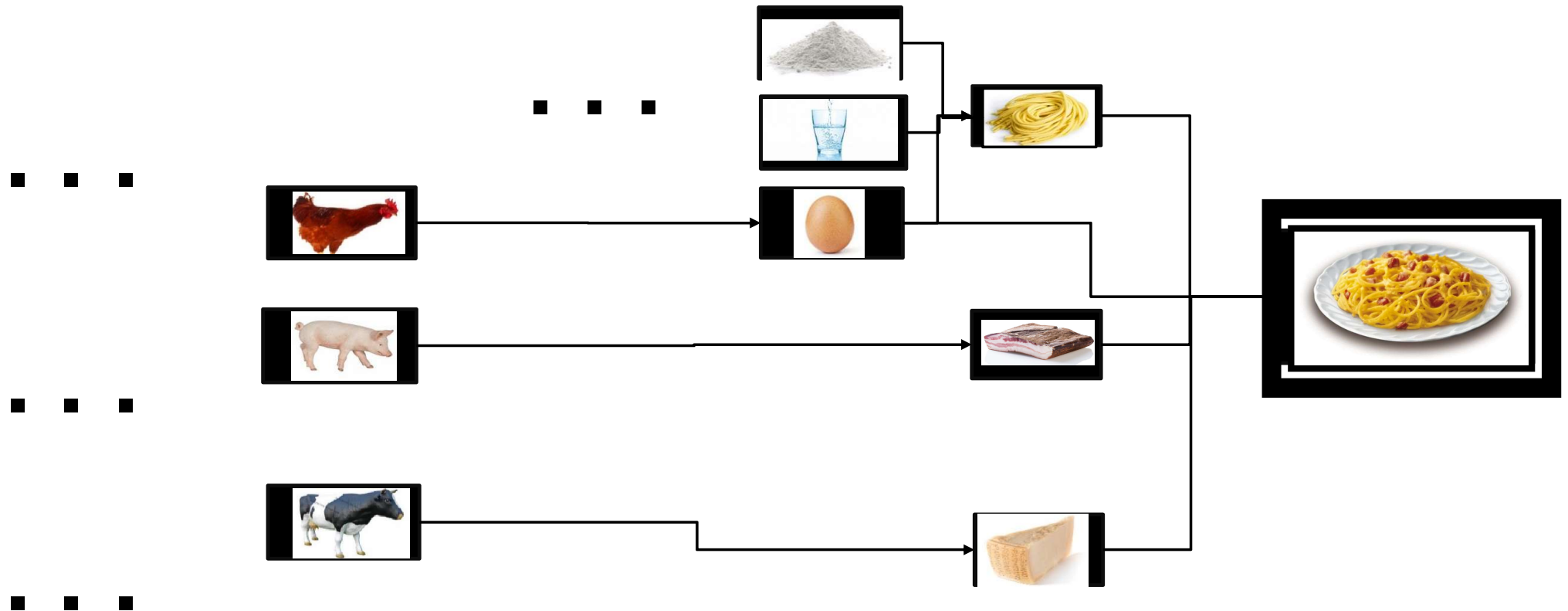
Metadata: Metadati definizioni e concetti base

- All **cooking recipes** are organized to describe some characteristics of the final product and its production process: preparation time, difficulty, ingredients and preparation methods.
- **Food labels** always have separate paragraphs describing the main characteristics, origin and certifications.
- Recipes and labels are two of the many metadata systems we know:
 - They tell us the meaning of the product
 - Help us use the product
 - They certify the origin or life cycle of the product.

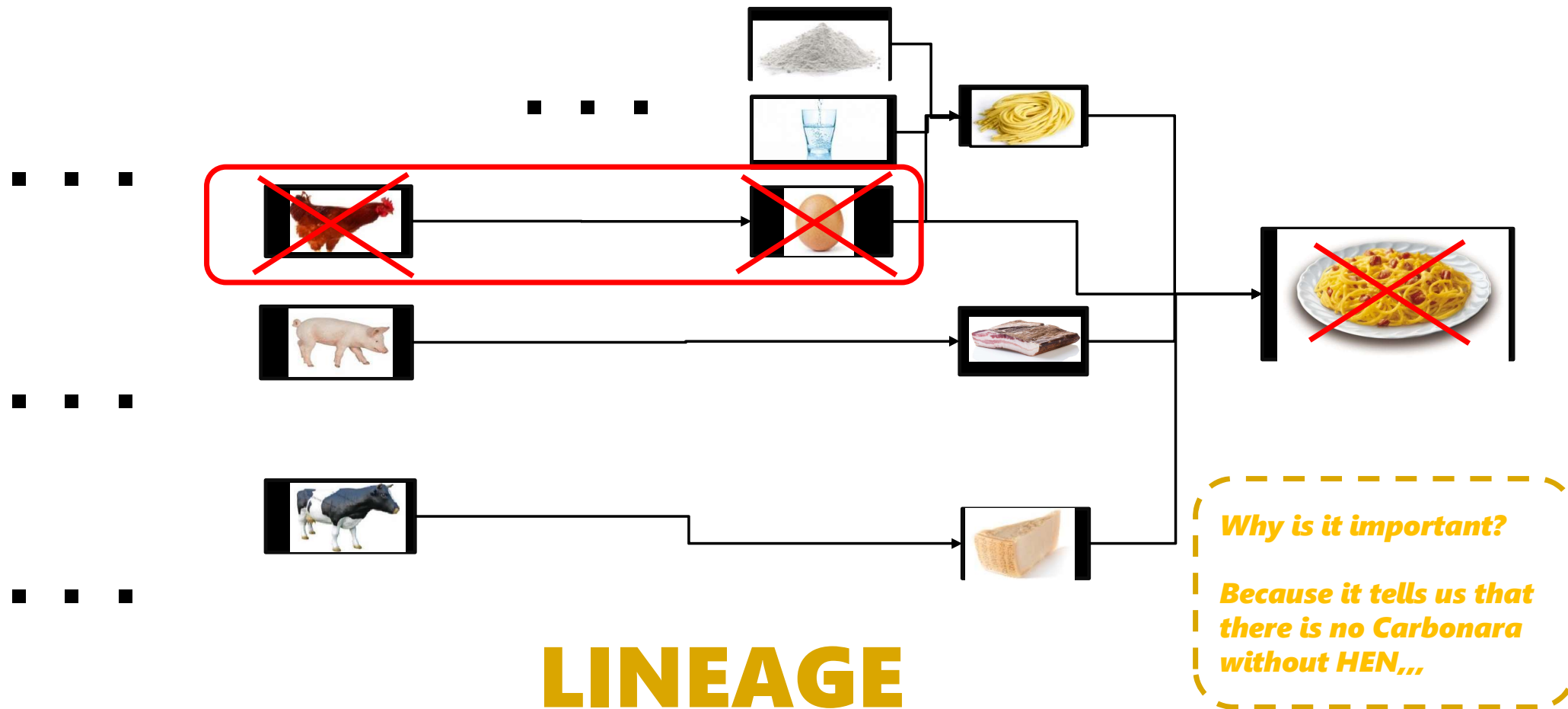
This **information is vital in our work**, where the data we use and process is increasingly numerous and heterogeneous.



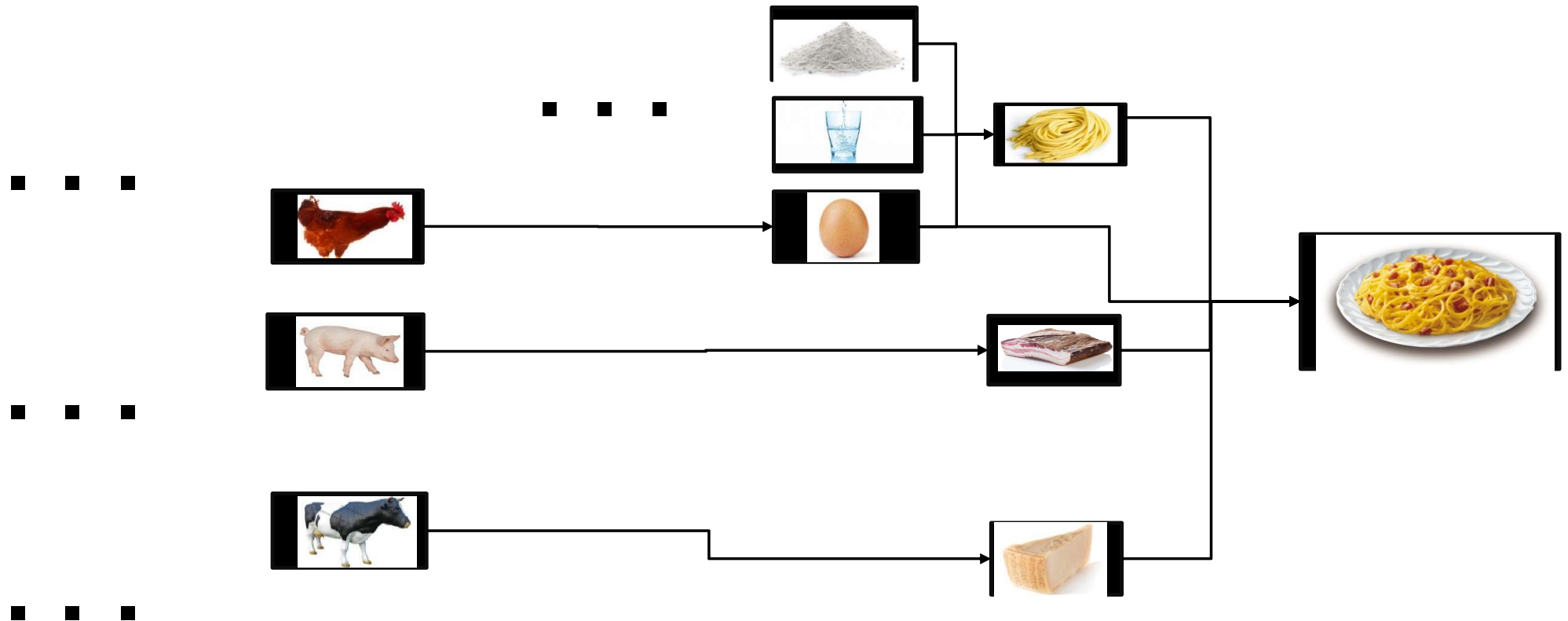
... genesis of a Carbonara...



... in the language of data governance experts...



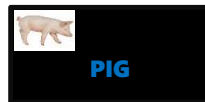
... dictionary of a Carbonara...



... dictionary of a Carbonara...



... in the language of data governance experts...

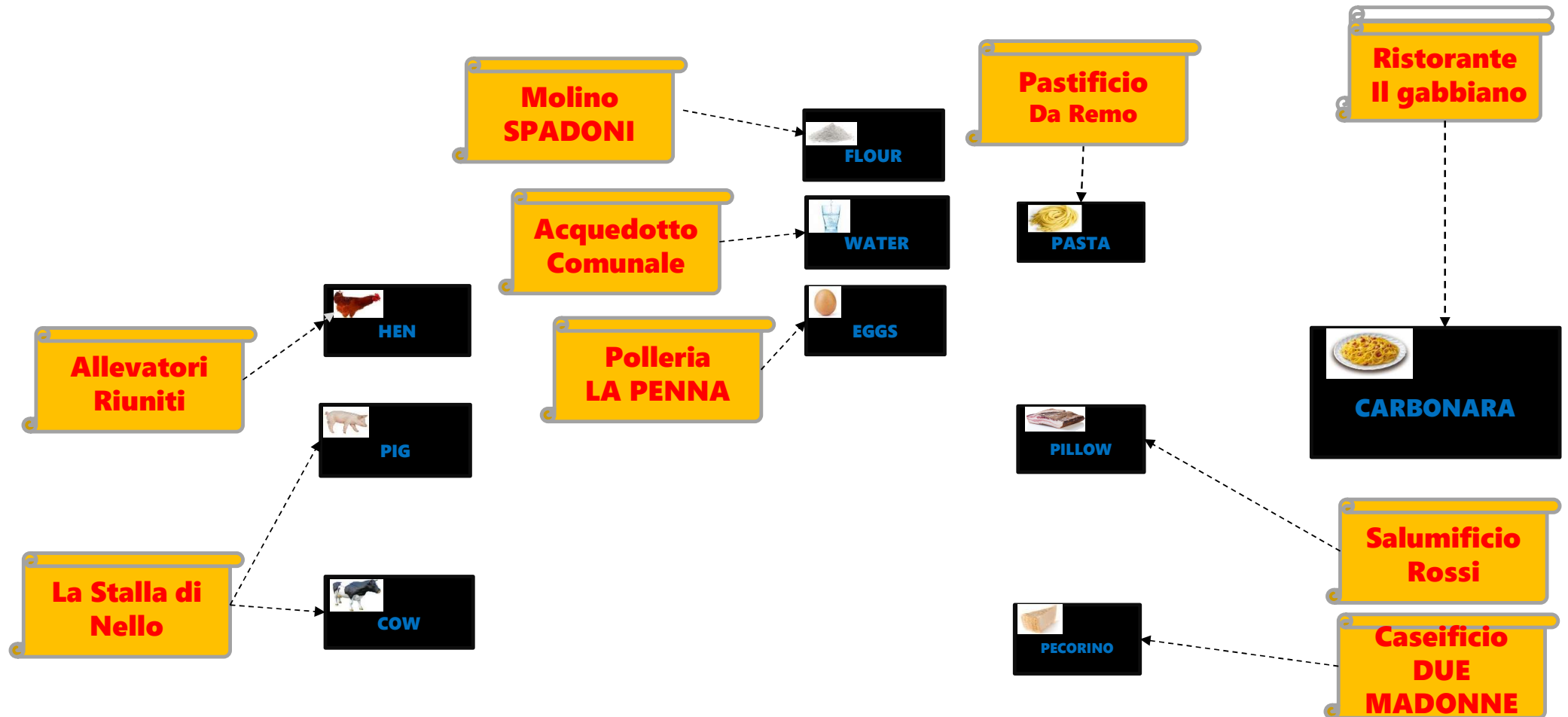


GLOSSARY

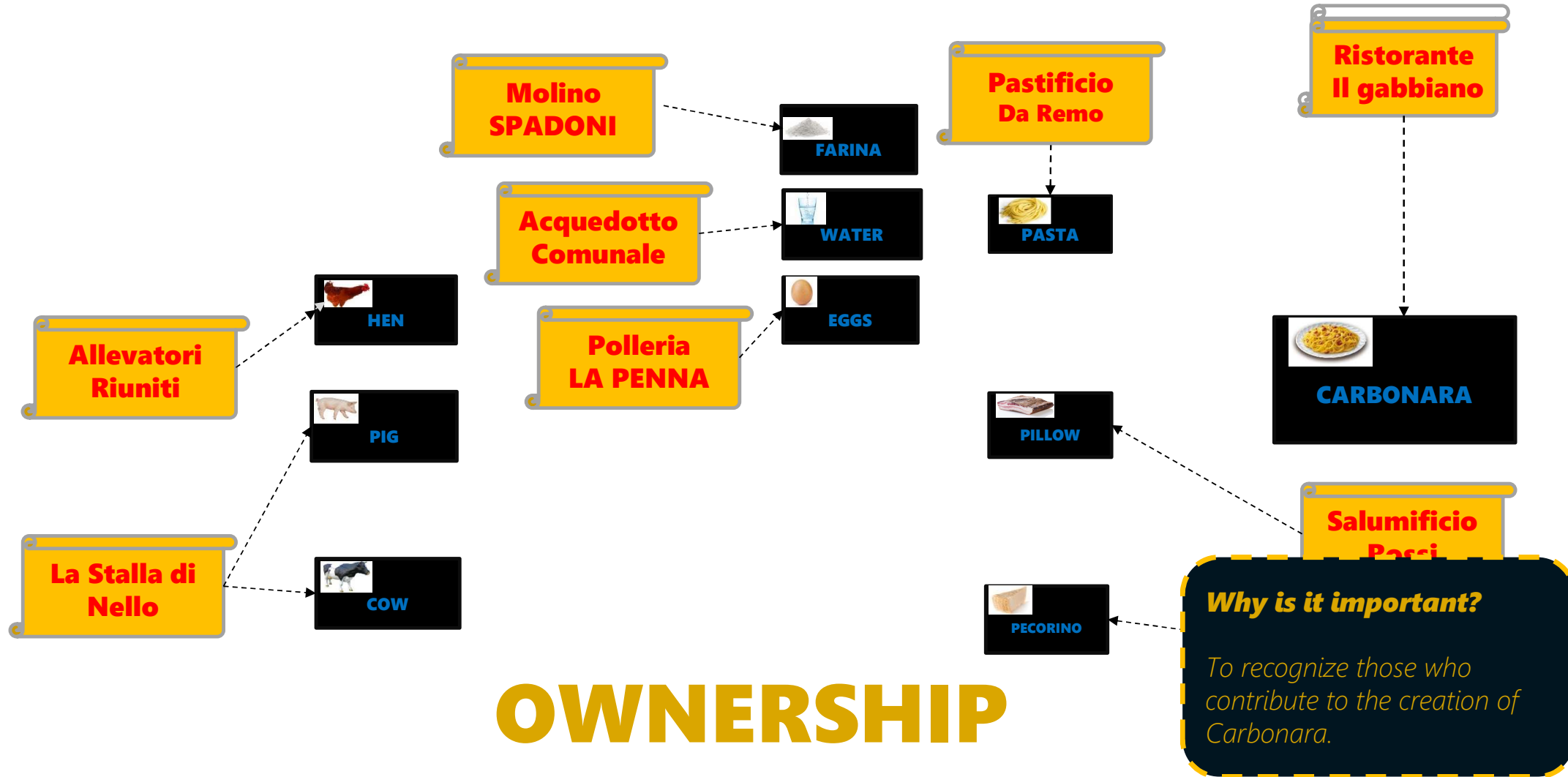
Why is it important?

Because it allows us to talk about Carbonara and what is around us, reducing ambiguity.

... the creators of Carbonara...



... in the language of data governance experts...



What it means to govern for data ?

